Stem Mixer · Quick Guide

FL Studio 21 · stock plugins · one template for Hip Hop, R&B and Afrobeats

1 · Every project, 3 steps

1. **File → New**, then drag all the pack's stems into the Playlist.
2. **Options → MIDI Settings** → click **loopMIDI Port** → **Controller type: Stem Mixer**.
   Already set to Stem Mixer? Switch it to *(generic controller)* and back to **Stem Mixer** to run it again.
3. Press **F9**. The mixer is built and every stem is already on the right track. **View → Script output** lists where each stem went.

Added more stems later? Repeat step 2. Your manual changes on track 24 and higher are never touched.

2 · The mixer (23 tracks)

Track	Name	Gets these stems (by file name)
1-7	KICK · SNARE/CLAP · HATS/PERC · DRUMS LOW · MID · HIGH · DRUM LOOP	kick · snare, clap, rim · hats, shaker, perc, congas, tops · "drums low/mid/high" · other drum loops
8-9	808 / BASS · BASS 2 / LOG DRUM	808, bass, sub · log drum
10-14	CHORDS/KEYS · MELODY 1-3 · FX/RISERS	chords, keys, piano, pads · everything melodic (spread over 3 tracks) · risers, impacts, sweeps
15-16	LEAD VOCAL · BACKING/ADLIBS	vocal, vox, hook, verse · adlibs, backing, harmonies, doubles, choir
17	FULL MIX (REF) · **muted**	full mix, instrumental, master. Unmute only to compare
18-21	**DRUMS BUS · BASS BUS · MUSIC BUS · VOCAL BUS**	The 4 groups. Each group's tracks feed into it, and it goes to the Master
22-23	REVERB · DELAY	Shared effects. You choose what goes in (section 3). FL 21 doesn't let scripts set send levels, so none are pre-made

You only ever touch **4 bus faders** to balance the song. The individual tracks are for fixing one sound.

3 · One-time setup: plugins on the template (about 15 min)

Do this **once** in a fresh project, then save it (section 5). FL presets can't be made outside FL, so this is the one manual step.

Track	Plugins, in order
DRUMS BUS	**Fruity Parametric EQ 2**: low-cut 30 Hz → **Fruity Compressor**: ratio 2:1, attack 30 ms, release 150 ms, threshold until it turns down 1-3 dB
BASS BUS	**Fruity Parametric EQ 2**: low-cut 25 Hz → **Fruity Limiter** for the sidechain (box below)
MUSIC BUS	**Fruity Parametric EQ 2**: low-cut 120 Hz, gentle dip around 300 Hz if muddy
VOCAL BUS	**Fruity Parametric EQ 2**: low-cut 90 Hz → **Fruity Compressor**: 3:1, fast attack, 3-6 dB of turning down
REVERB	**Fruity Reeverb 2**: Dry 0%, Wet 100% → **Fruity Parametric EQ 2**: low-cut 250 Hz, high-cut 9 kHz → **Fruity Limiter** for the vocal duck (box below)
DELAY	**Fruity Delay 3**: Dry 0%, time 1/4 beat, low feedback → **Fruity Parametric EQ 2**: low-cut 300 Hz, high-cut 7 kHz

Sending a track to REVERB or DELAY: select the track (e.g. LEAD VOCAL) → **left-click** the small arrow under REVERB → turn the little send knob on that arrow down to about **-10 dB**, then raise to taste. Good candidates: lead vocal, backing, snare, chords, melodies.

The only 2 sidechain links you need (select the trigger track → right-click the small arrow under the target track → *Sidechain to this track*):

1. **KICK → BASS BUS.** In that Fruity Limiter, turn CEIL up, switch to **COMP**, right-click **SIDECHAIN** → KICK, ratio 4:1, lower the threshold until the bass dips on each kick. No kick stem? Use **DRUMS LOW**.
2. **LEAD VOCAL → REVERB.** Same Limiter steps (trigger: LEAD VOCAL). The reverb dips while the vocal sings, so words stay clear.

4 · Genre settings (mix)

	Hip Hop	R&B	Afrobeats
Kick → bass dip	7-10 dB, hard	3-5 dB, smooth	4-6 dB (bass + log drum bounce)
808 / bass	Add **Fruity Soft Clipper** on 808 track, so it's heard on phones	Clean, no clipper	Light Soft Clipper on LOG DRUM
Lead vocal EQ (on LEAD VOCAL track)	+2 dB at 3-5 kHz, cut 200-400 Hz if muddy	Cut 250-400 Hz, +2 dB "air" shelf at 10-12 kHz	Cut 200-400 Hz, gentle +2 dB at 2-4 kHz, **+2-4 dB above 10 kHz**
Vocal comp	4:1, 4-6 dB	2:1, 2-3 dB (smooth)	3:1, 4-6 dB on the hook
REVERB size	Short, ~1 s, low level (dry, upfront)	Long, 2-2.5 s, more level	Short plate, 0.8-1.5 s
DELAY preset (Fruity Delay 3)	HIP HOP - Adlib 1-8 · HIP HOP - Vocal Throw 1-4	RnB - Dreamy 1-4 dotted	AFROBEATS - Last Word Throw 1-4 · AFROBEATS - Bounce 1-8 dotted
Backing vs lead	Adlibs ~6 dB under	Harmonies 6-9 dB under	Doubles 4-6 dB under, harmonies 6-9 dB under

Delay presets (installed on your PC). Put **Fruity Delay 3** on the **DELAY** track → click **Presets** at the top of the plugin → pick one. They are all **tempo-synced** (they follow your BPM automatically), with dry 0% and darker repeats so they sit behind the vocal.

Preset	Time	Repeats (feedback)	Tone	Use it for
HIP HOP - Adlib 1-8	1/8	25%	6 kHz	Fast echo on adlibs, rap line ends
HIP HOP - Vocal Throw 1-4	1/4	20%	5 kHz	Echo on single words/hooks
AFROBEATS - Last Word Throw 1-4	1/4	12% (one clean repeat)	6 kHz	Last word of each phrase
AFROBEATS - Bounce 1-8 dotted	1/8 dotted	30%, ping pong	7 kHz	Groove on plucks, guitar, chants
RnB - Dreamy 1-4 dotted	1/4 dotted	35%, ping pong	4 kHz (warm)	Lush vocals, pads

Tweak by ear: more FEEDBACK = more repeats, lower CUTOFF = darker. To save your tweak: plugin menu (top-left arrow) → Save preset as.... Manual maths if ever needed: 1/4 note in ms = 60,000 ÷ BPM (86 BPM = 698 ms).

Balancing order (all genres)

1. Pull the 4 bus faders until each **peaks around -6 dB** and the Master peaks around **-6 to -3 dB** before mastering.
2. Kick + bass first → rest of drums → **lead vocal 3-6 dB above the loudest instrument** → music → FX.
3. Bass tracks: turn the track's stereo separation knob fully to **merged** (mono).

5 · Mastering presets (already installed on your PC)

Load one in 2 clicks: in the Mixer, right-click the **Master** track name → **File** → pick **MASTER - Afrobeats**, **MASTER - Hip Hop** or **MASTER - RnB**. Then raise the **Fruity Limiter GAIN** until you reach the loudness target.

#	Plugin (Master slots 1-4)	MASTER - Hip Hop	MASTER - RnB	MASTER - Afrobeats
1	**Fruity Parametric EQ 2**	FL preset *"20Hz + 18kHz cut"*: removes rumble and harsh ultra-highs		
2	**Fruity Compressor**	FL preset *"Master 1"*: 3.2:1, attack 11 ms, punchy control	FL preset *"Light"*: 1.7:1, slow 49 ms attack, gentle glue that keeps drum hits snappy	
3	**Fruity Soft Clipper**	On: shaves peaks for loudness	**Off** (smoother, more dynamic)	On
4	**Fruity Limiter**	CEIL **-1.0 dB** (safe for streaming). **You set GAIN** per song, since it depends on how loud your mix is		
	Loudness target	**-9 to -7 LUFS**	**-11 to -9 LUFS**	**-9 to -8 LUFS**

The compressor's "threshold" decides how hard it works. Play the loudest part and aim for **1-2 dB** of turning down. If it's more, raise THRESHOLD a little. Tweaked a preset you like? Right-click Master → File → **Save mixer track state as...** to save your own version. Backup copies of the 3 presets are in the ZIP (`Mixer presets` folder).

> **Measuring loudness:** FL 21 has no simple loudness (LUFS) meter built in. Add the free **Youlean Loudness Meter** as the very last plugin on the Master. It only measures. Keep **True Peak ≤ -1 dB**.

Final checks

- Unmute track 17 (FULL MIX REF) and compare. Are your low end and vocal as clear?
- Listen on phone speakers and earphones. Is the 808 / log drum still audible?
- Switch the whole Master chain off and on. It should sound better, not just louder.

6 · Optional: save the bus plugins as a template

The script rebuilds the mixer every time, so you don't need a template. To skip redoing section 3 each time: after adding those plugins in an empty project (no stems), **File → Save as...** → `Stem Mixer Template.flp`. Open it, drag stems in, then run Stem Mixer. It keeps your plugins and just routes the stems.

If something's off: View → Script output shows what the script did. A stem on the wrong track? In the Channel Rack, change its mixer track number by hand. If the script ran on a project with its own named mixer tracks, it skips it on purpose (use File → New).
Sources: Image-Line MIDI scripting manual · FL Studio API stubs · Mix Elite (sidechain in FL Studio) · Loopmasters (routing & busses) · Avid (mixing drums) · Antares (Afrobeats vocal production) · Venia Mastering & Mastering France (hip-hop loudness) · Image-Line forum (clip before limit) · The Vocal Market (vocal levels). Numbers are starting points. Trust your ears.